

Operations/Work/Job Activity covered by this assessment: High Velocity Oxy-Fuel (HVOF Spray)

Required Personal Protective Equipment (PPE) :

Safety glasses must be worn at times.



Long and loose hair must be contained.



Appropriate footwear with substantial uppers must be worn.



Close fitting/protective clothing to cover arms and legs must be worn.



Respiratory protection devices required.



Oil free leather gloves and spats must be worn.



Hearing protection must be used when using this machine.

Loading & Unloading Check:

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling.

Operations Procedure:

- 1) Refer Pre-Inspection Work Instructions PSP/WI/12 to pre-inspect the Component.
- 2) Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
- 3) Mask with High Temperature Adhesive Tape/Metal Shim/ to protect unwanted area from over spraying during HVOF spray coating.
- 4) Load the Component in the acoustic room and clamp it thoroughly on the chuck with proper packing to prevent clamping marks on the Component.
- 5) Start Exhaust Fans before starting the HP/HVOF Control Unit.
- 6) Open valve of Nitrogen and Oxygen Cylinders and set the pressure according to the parameter mentioned in the Technical Bulletin.
- 7) Start the JP8000 control unit with your Login ID.
- 8) Check Oxygen, Fuel & Water Flow & Pressure in the manual mode screen.
- 9) Check Combustion Pressure (psi).
- 10) Select the required powder recipe from the recipe menu w.r.t the Job Card.
- 11) Pre-heat the powder in the powder feeder hopper and maintain the temperature with the heating blanket attached.
- 12) Measure the powder flow rate (75 to 80 grams/min) as per parameter and set the Powder feeder RPM.
- 13) Set the job speed and Robot program speed, refer Job Card or instructions will be given by Supervisor/WM.

PLASMA SPRAY PROCESSORS Works Instruction	QF/QMS/02/07	
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- 14) Closed the Acoustic Door and Wait for signal before igniting the JP Torch. Go to Development screen and press the gun ignition. Wait for the gun to ramp up till you get the desire combustion pressure as per the parameter.
- 15) Pre-heat the skin temperature of the job up to 45°C to 60°C, check with pyrometer.
- 16) "ON" Powder feeder and wait till the powder is stabilize and spray the job layer by layers to get a uniform thickness all over.
- 17) Intermediate dimension should be checked to control dimension as per job card.

Shutdown :

1. Shut the Powder Feeder and wait for the feed lines to clear.
2. Shut down the ignition flame and bring the Robot to its home position.
3. Cool the job gradually and recheck the dimension. Fill up the Job Card as per the data and Tag it.
4. Refer to Final Inspection Work Instruction PSP/WI/10 and get the approval Supervisor.

Assessor :		Signature:		Date:	
Approved:		Signature:		Date:	